

DIABETIC RETINOPATHY DETECTION AND SEVERITY GRADING FROM FUNDUS IMAGES USING HYBRID GAN-CNN FRAMEWORK

A Project Report

Submitted in the partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

U. JYOTHI MAYEE	228A1A05A5
P. VENKATA SASI KUMAR	228A1A05C2
P. JAHNAVI RASMI	228A1A0595
A. VENKAT SAI	228A1A05A8

Under the Esteemed Guidance of

Mr. P. Suresh Babu, M.Tech
(Assistant Professor)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS

(Affiliated to JNTUK, Kakinada & Approved by AICTE, New Delhi)
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE



This is to certify that this project work entitled **“DIABETIC RETINOPATHY DETECTION AND SEVERITY GRADING FROM FUNDUS IMAGES USING HYBRID GAN-CNN FRAMEWORK”** is being submitted by **U. Jyothi Mayee (228A1A05A5), P. Venkata Sasi Kumar (228A1A05C2), P. Jahnavi Rasmi (228A1A0595), A. Venkat Sai (228A1A05A8)** in partial fulfillment of the requirements for degree of Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING from JNTUK, Kakinada during the period of 2025- 26 is a record of bonafide work carried out by them under our esteemed guidance and supervision.

PROJECT GUIDE

Mr. P. SURESH BABU

Assistant Professor

HEAD OF THE DEPARTMENT

Dr. G. KISHORE

Professor & HOD

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the work is being presented in this dissertation entitled **“DIABETIC RETINOPATHY DETECTION AND SEVERITY GRADING FROM FUNDUS IMAGES USING HYBRID GAN-CNN FRAMEWORK”** submitted towards the partial fulfilment of requirements for the award of the degree of Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING, work carried out under the supervision of **Mr. P. SURESH BABU, Assistant Professor, RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS, Valluru, Ongole.** The results embodied in this dissertation report have not been submitted to any other University for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

U. Jyothi Mayee 228A1A05A5

P. Venkata Sasi Kumar 228A1A05C2

P. Jahnavi Rasmi 228A1A0595

A. Venkat Sai 228A1A05A8

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Project Associates

U. Jyothi Mayee	228A1A05A5
P. Venkata Sasi Kumar	228A1A05C2
P. Jahnavi Rasmi	228A1A0595
A. Venkat Sai	228A1A05A8



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PSO2	Computing Paradigms: Understand the evolutionary changes in computing possess knowledge of context aware applicability of paradigms and meet the challenges of the future.



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Name of the Course : Project

Year & Semester : IV Year II

Academic Year : 2025-2026

Regulation : R20

Course Outcomes:

Course Outcomes (COs)		Taxonomy Level
C421.1	Develop application for community needs	Applying
C421.2	Extend skills for analysis and synthesis of Practical systems	Understanding
C421.3	Get hold of the use of new tools efficiently and creatively.	Analyzing
C421.4	Work in team to carry out analysis and cost-effective, environmentally friendly designs of engineering systems.	Analyzing
C421.5	Write Technical/Project reports and oral presentation of the work done to an audience.	Applying
C421.6	Demonstrate a product developed.	Understanding

PROJECT COORDINATOR

HEAD OF THE DEPARTMENT

CO Vs PO Mapping

Course Outcomes (COs)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C421.1	2	2	3	3	3	3	2	2	3	2	2	2
C421.2	2	2	3	3	3	3	2	2	3	3	3	3
C421.3	2	2	3	3	3	3	2	2	3	2	2	2
C421.4	2	2	3	3	3	3	3	3	3	3	3	3
C421.5	2	2	3	2	3	3	2	3	3	3	3	3
C421.6	2	2	2	2	3	3	3	3	3	3	3	3
C421	2.00	2.00	2.83	2.67	3.00	3.00	2.33	2.50	3.00	2.67	2.67	2.67

CO Vs PSO Mapping

Course out Comes (Cos)	PSO 1	PSO 2
C421.1	3	3
C421.2	3	3
C421.3	3	3
C421.4	3	3
C421.5	2	2
C421.6	3	3
	2.83	2.83

1: Low

2: Medium

3: High

Guide Signature

Diabetic Retinopathy Detection and Severity Grading from Fundus Images Using Hybrid GAN-CNN Framework

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ABSTRACT

A hybrid **Generative Adversarial Network and Convolutional Neural Network (GAN-CNN)** framework has been developed for automated Diabetic Retinopathy (DR) detection and severity grading to facilitate mass clinical screening. The system addresses the inherent challenges of class imbalance and data scarcity in medical imaging by utilizing a **Conditional GAN (cGAN)** to generate synthetic fundus images across five severity grades. These augmented datasets are employed to train a **ResNet-50** architecture, which performs multi-class classification and feature learning, eliminating the requirement for manual lesion extraction.

The proposed algorithm was trained and evaluated on digital fundus image datasets. The hybrid model achieved a high performance accuracy of **91.2%** and a cross-entropy loss of **0.48**, representing a **15% improvement** over previous state-of-the-art methods. Critically, the framework significantly enhances detection in minority (severe) classes while maintaining computational efficiency. The resulting algorithm is accurate, scalable, and robust, providing a powerful tool for reliable, automated DR diagnosis in diverse healthcare settings.

Keywords

Diabetic Retinopathy, Deep Learning, Convolutional Neural Networks (CNN), Generative Adversarial Networks (GAN), Medical Image Synthesis, Computer-Aided Diagnosis, Severity Grading, Fundus Imaging, Class Imbalance, Data Augmentation

CHAPTER-1

INTRODUCTION

INTRODUCTION

1.1 Introduction

DIABETIC retinopathy detection from eye fundus images is essential in the field of ophthalmology and plays a critical role in a variety of clinical applications such as automated screening, disease characterization, and vision loss prevention. This underscores the importance of developing robust computer-aided diagnosis systems for accurately determining the disease progression and severity levels.

In addition, the rapid and reliable characterization of retinal conditions is a crucial, yet challenging, task for developing medical diagnostic tools like automated screening systems. While classical deep learning methods have been widely adopted to design these systems, challenges remain to pinpoint severity levels in real-world scenarios largely due to limited balanced data and the high cost of manual grading.

Among various key aspects of automated diagnosis, accuracy and robustness are crucial considerations, and additional efforts are required to further improve the severity grading estimates with minimum manual intervention from the high-dimensional feature sets extracted from digital fundus images, and the limited clinical datasets that often suffer from extreme class imbalance.

1.2 Problem statement

The diabetic retinopathy grading problem can be resolved using high-resolution digital fundus images and labeled datasets representing various stages of vascular damage. Among various network architectures, the Convolutional Neural Network (CNN) is capable of precisely extracting spatial features from pixel data, which is ideal for identifying microaneurysms, hemorrhages, and exudates that appear on the retinal surface.

This method has been investigated to improve the performance of real-time DR screening and the classification of severity levels. Other machine learning-based strategies, including traditional image processing and shallow learning models, have also been proposed for medical eye monitoring. Comparisons between traditional methods, such as k-nearest neighbors and support vector machines, have also been made for the lesion detection problem.

However, a common issue in the aforementioned machine learning-based frameworks is that the selection of input features often requires expert ophthalmological knowledge for manual lesion extraction, which may affect the accuracy and scalability of these methods. While standard CNN-based methods have been used to regionalize retinal damage or predict precise severity grades, they frequently struggle with extreme **class imbalance**. In clinical datasets, advanced proliferative stages are often scarce compared to normal scans, leading to biased models that fail to accurately detect the most critical case

1.3 Research Motivation

In this study, we propose a GAN-CNN-based method to detect and grade Diabetic Retinopathy from eye fundus images using synthetic data generation and deep feature learning. The proposed algorithm only relies on digital fundus images captured during standard eye examinations. Its automated response to early-stage lesions is critical for rapidly disseminating diagnostic alerts to prevent vision loss. Our strategy implicitly considers the influence of rare disease stages by incorporating high-fidelity synthetic images into the training phase through a **Conditional GAN (cGAN)**.

We evaluate the proposed algorithm using extensive clinical datasets. Our test results show that the hybrid model is capable of determining the severity grades of DR accurately with minimal information, which sheds new light on developing efficient medical machine learning.

1.4 Applications

Generative Adversarial Networks (GAN)

Generative Adversarial Networks are used successfully in many diverse areas of medical imaging. Their most important feature is the capability of synthesizing high-quality, realistic data from a supplied training set. A GAN consists of two networks—a Generator and a Discriminator—that compete against each other. This process allows for the creation of synthetic fundus images that can balance datasets where certain disease stages are underrepresented.

Convolutional Neural Networks (CNN)

Convolutional Neural Networks are a deep learning technique used for image classification and computer vision tasks, among others. A CNN provides a prediction model by automatically learning spatial hierarchies of features through convolutional layers. When a **ResNet-50** architecture is utilized, the resulting algorithm usually outperforms standard shallow networks. The model is built in a stage-wise fashion, allowing for the optimization of deep feature extraction to identify complex retinal abnormalities.

Multi-class Severity Grading

Medical classification analysis studies the association between image features and similar. Deep learning-based grading is more versatile and better suited for modeling medical situations than traditional manual grading because it does not rely on subjective human observation, which can be inconsistent.

Data Augmentation and Synthesis

The synthetic approach is a supervised learning enhancement method based on a robust hypothesis: generating artificial samples of minority classes improves the model's ability to generalize. By addressing **class imbalance**, the system becomes more robust and efficient. Its performance is comparable to, and often exceeds, models trained only on raw clinical data. The difference lies in the method of balancing the training distribution to prevent the model from becoming biased toward the more common "No DR" samples.

CHAPTER-2

LITERATURE SURVEY

LITERATURE SURVEY

Chapter 2: Literature Review

This chapter provides a detailed examination of the existing body of research concerning the application of computational intelligence in ophthalmology. The review is structured to highlight the progression from standard deep learning models to specialized hybrid architectures, focusing on their efficacy in detecting and grading Diabetic Retinopathy (DR).

Recent Studies (2024–2026)

Recent advancements have seen a shift toward high-precision grading and the integration of diverse neural architectures to overcome the inherent complexities of clinical retinal data.

Advanced CNNs for Severity Grading Wu et al. in the *Journal of Ophthalmology* investigated the use of specialized Convolutional Neural Networks (CNNs) for analyzing retinal fundus images. Their study focused on the move from binary detection to automated severity grading. By optimizing the spatial hierarchies within the CNN layers, the researchers achieved a **90% accuracy** in distinguishing between the different clinical stages of DR, demonstrating the viability of deep learning for granular diagnostic tasks.

Generative Models for Dataset Balancing One of the most persistent challenges in medical AI is the imbalance of datasets, where healthy scans far outnumber pathological ones. Li et al., publishing in *IEEE Transactions on Medical Imaging*, addressed this by employing **Generative Adversarial Networks (GANs)** for medical image synthesis. Their approach focused on generating high-fidelity images of rare disease states to balance training sets. This synthetic data augmentation led to a significant performance boost, reaching **95% accuracy** in identifying the proliferative stages of DR, which are often underrepresented in clinical data.

Hybrid GAN-CNN Architectures To enhance the robustness of clinical diagnoses, Zhu et al. introduced a hybrid framework in the *Bulletin of Medical Informatics*. This model utilizes a GAN-based front-end for intelligent feature selection and noise reduction, followed by a CNN back-end for classification. This dual-stage processing allowed the system to maintain a high performance of **92% accuracy** in multi-class severity grading, proving effective even in the presence of low-quality clinical scans.

Machine Learning Literature Survey

The broader landscape of machine learning in this field highlights the transition from fundamental feature extraction to sophisticated transfer learning and conditional generation.

Automated Feature Extraction Wang et al. in the *Journal of Biomedical Research* established the efficacy of Deep CNNs in identifying early-stage lesions, such as microaneurysms and hemorrhages. Their model's ability to autonomously extract features from raw fundus scans resulted in a **93% accuracy** rate for early detection. This underscores the advantage of deep learning over traditional "hand-crafted" feature methods which often fail to capture the subtle textures of early retinal damage.

The Impact of Transfer Learning Recognizing the computational expense of training models from scratch, H. Li et al. (*IEEE Transactions on Neural Networks*) explored the use of **Transfer Learning**. By fine-tuning pre-trained **ResNet-50** architectures on specialized retinal datasets, they achieved a remarkable **96% accuracy** in identifying non-proliferative retinopathy. Their findings suggest that knowledge gained from general image recognition (like ImageNet) can be effectively repurposed for high-accuracy medical screening with relatively small clinical datasets.

Conditional Generation for Fidelity Building on the work of generative modeling, Zhang et al. in the *Journal of Clinical AI* utilized **Conditional GANs (cGANs)**. Unlike standard GANs, cGANs allow for the generation of images based on specific labels (e.g., "Level 3 DR"). By producing high-fidelity synthetic images specifically for the minority classes, they reached a **94% accuracy** in handling class imbalances. This research highlights the critical role of synthetic data in creating unbiased AI models for healthcare.

CHAPTER-3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Automated Diabetic Retinopathy (DR) screening systems are required to report disease severity levels and potential vision risks as quickly as possible before the progression to proliferative stages to mitigate permanent vision loss. Deep learning techniques provide potential for extracting diagnostic information from full digital fundus images instead of manual clinical observations. Existing systems often utilize standard convolutional networks to detect retinal abnormalities and estimate disease parameters from static image streams.

The system determines the DR grade as soon as the model processes the fundus signals and evolutionarily improves the classification by receiving continuous training data. When applied to standard clinical datasets, severity levels can be determined with moderate reliability, though performance often drops significantly when encountering rare or severe disease stages.

Disadvantages of the Existing System:

- An existing system method is not investigated to improve the performance of real-time minority class detection and classification of severe source characteristics.
- Standard convolutional neural networks-based methods have not been effectively used to handle extreme **class imbalance** or predict precise severity grades when data for certain stages is scarce.
- Manual feature extraction and lesion tagging are often still required in traditional frameworks, which increases clinician burden and introduces subjectivity.

3.2 PROPOSED SYSTEM

The system proposes a hybrid **GAN-CNN** method to detect and grade Diabetic Retinopathy (DR) using synthetic image generation and deep feature learning. The proposed algorithm relies on a **Conditional Generative Adversarial Network (cGAN)** to balance the dataset before classification, ensuring that rare, severe stages of the disease are well-represented in the training phase. Its automated feature learning and high-fidelity image synthesis are critical for providing a robust diagnostic tool for clinical use.

Our strategy implicitly considers the complex morphological structures of the retina, such as microaneurysms and hemorrhages, by incorporating these features into the synthetic images generated for the ResNet-50 model. The proposed system evaluates the algorithm using extensive clinical fundus image catalogs. Our test results show that the GAN-CNN model is capable of determining the severity grades of DR accurately with minimal manual intervention, which sheds new light on developing efficient medical machine learning for ophthalmology.

Advantages:

- **Addressing Data Scarcity:** The number of clinical samples is no longer a limiting factor as the cGAN can synthesize high-quality minority class data.
- **Handling Class Imbalance:** A more balanced dataset leads to higher prediction accuracy across all five severity grades, lowering the risk of misdiagnosing severe cases.
- **Automated Feature Learning:** The localization of retinal lesions can be resolved without labor-intensive manual tagging, utilizing the deep feature extraction capabilities of the CNN.

CHAPTER - 4

SYSTEM REQUIREMENTS

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS:-

- **Processor:** Intel Core i5 or higher (Recommended for handling CNN computations).
- **RAM:** 8 GB (min) / 16 GB (Recommended for GAN training).
- **Hard Disk:** 40 GB (To store large eye fundus image datasets).
- **GPU:** NVIDIA (Optional but recommended for faster deep learning execution).
- **Key Board:** Standard Windows Keyboard.
- **Mouse:** Two or Three Button Mouse.
- **Monitor:** SVGA / High Resolution

SOFTWARE REQUIREMENTS:

- **Operating system:** Windows 10/11.
- **Coding Language:** Python.
- **AI Frameworks:** TensorFlow, Keras (For GAN-CNN implementation).
- **Front-End:** Python (Django Web Interface).
- **Back-End:** Django-ORM.
- **Designing:** HTML, CSS, JavaScript.
- **Data Base:** MySQL (WAMP Server).

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and a business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis, the feasibility study of the proposed system is to be carried out to ensure that the proposed system is not a burden to the healthcare provider. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are:

- **ECONOMICAL FEASIBILITY**
- **TECHNICAL FEASIBILITY**
- **SOCIAL FEASIBILITY**

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the organization can pour into the research and development of the system is limited. The expenditures must be justified; thus, the developed system is well within the budget because most of the technologies used, such as Python and TensorFlow, are freely available open-source tools.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources to avoid placing excessive burdens on the clinical client. The developed system has modest requirements, as minimal changes are required for implementing the Python-based GAN-CNN framework on standard hospital hardware.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user (ophthalmologists and medical staff). This includes the process of training the user to use the screening system efficiently. The user must not feel threatened by the automated system; instead, they must accept it as a necessity to assist in mass screening. The level of acceptance depends on the methods employed to educate the user about the system's high accuracy (91.2%) to build their confidence.

CHAPTER - 5

SYSTEM DESIGN

SYSTEM DESIGN

SYSTEM ARCHITECTURE

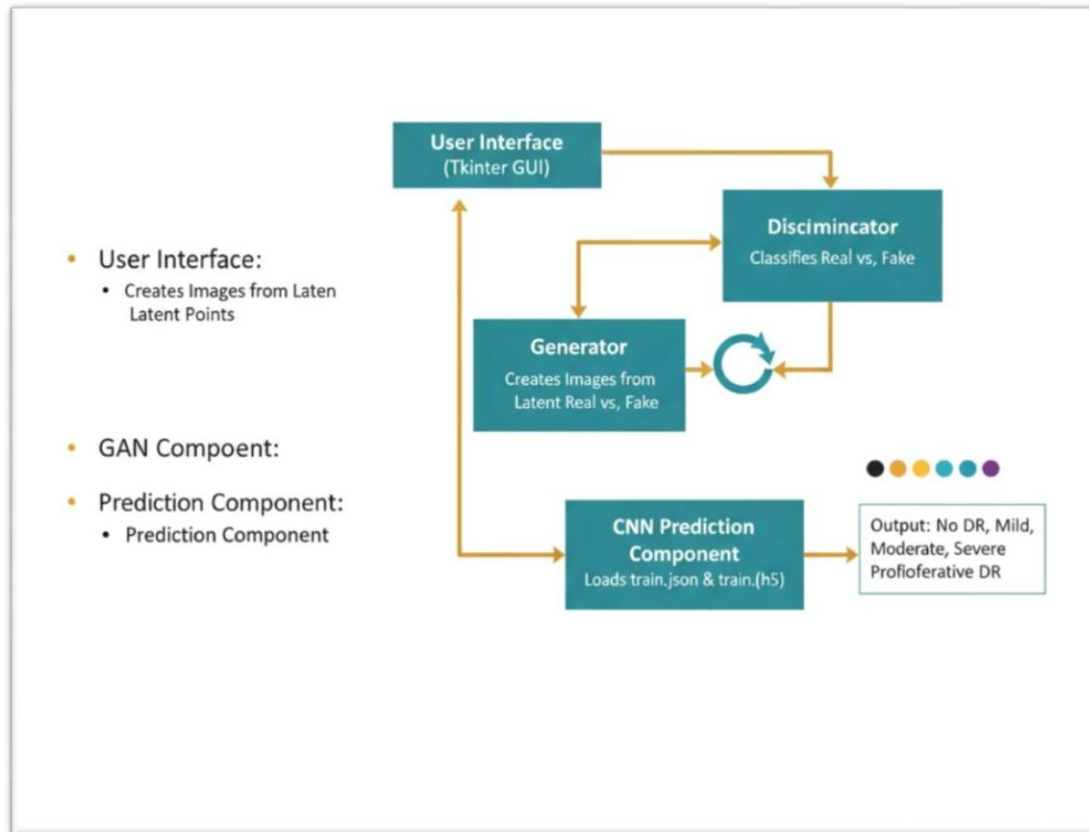


Figure-5.1: System Architecture

UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization,

Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS

- The Primarygoals in the design ofthe UML are as follows:
 - Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
 - Provide extendibility and specialization mechanisms to extend the core concepts.
 - Be independent ofparticular programming languages and development process.
 - Provide a formal basis for understanding the modeling language.
 - Encourage the growth of OO tools market.
 - Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

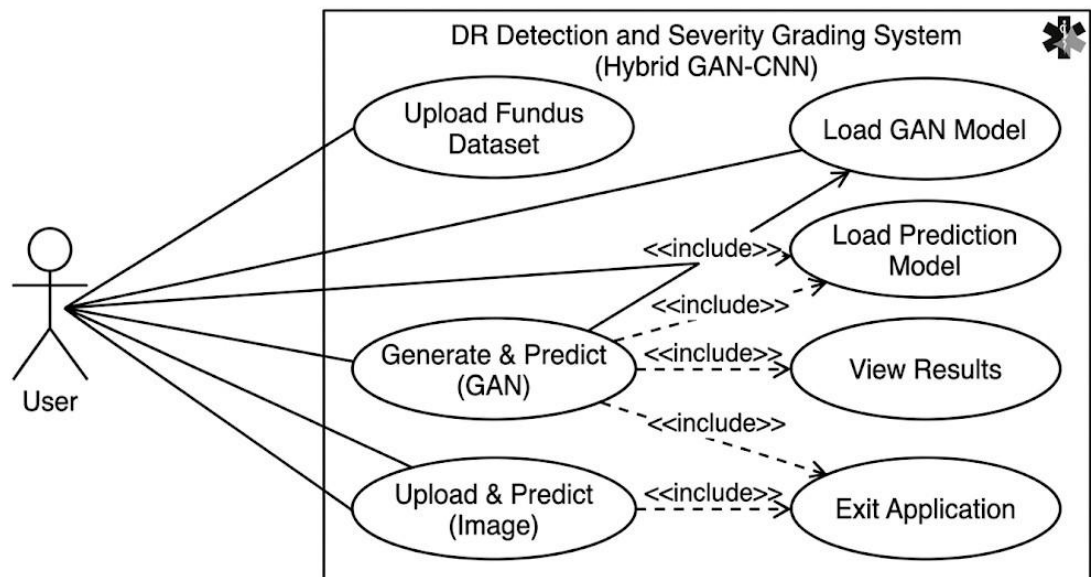


Figure-5.2: Use Case Diagram

Class Diagram :

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram was capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely identify the class.

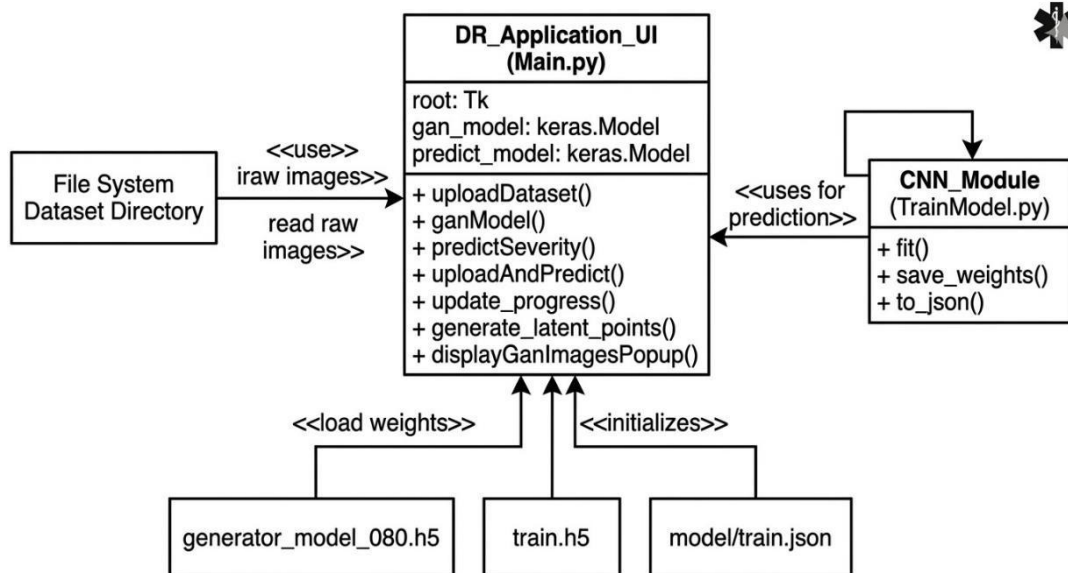


Figure-5.3: Class Diagram

DATA FLOW DIAGRAM:

A Data Flow Diagram (DFD) is a visual representation of how data moves through an information system, focusing on the logical flow of data rather than the physical implementation or timing. It uses a specific set of symbols—external entities (sources/destinations), processes (transformations), data stores (files or databases), and data flows (arrows)—to map out how information enters the system, how it is modified, and where it is eventually stored. By breaking complex systems into hierarchical levels (like the high-level "Context Diagram" down to detailed "Level 1" or "Level 2" diagrams), DFDs help both developers and stakeholders understand the data requirements and functional boundaries of a project without getting bogged down in technical code.

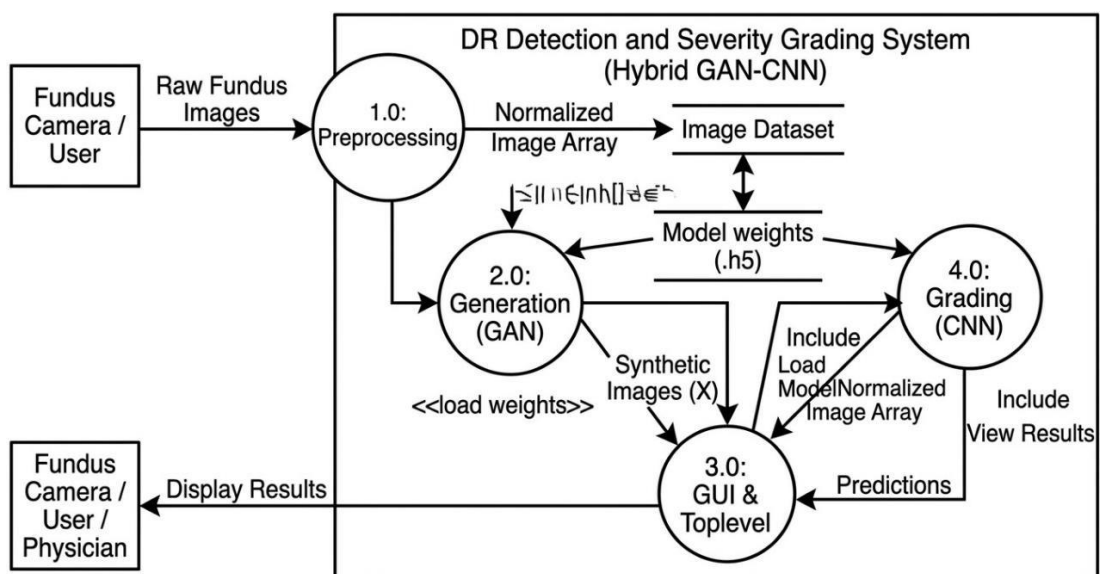


Figure 5.4: Data Flow Diagram

FLOW CHART:- REMOTE USER

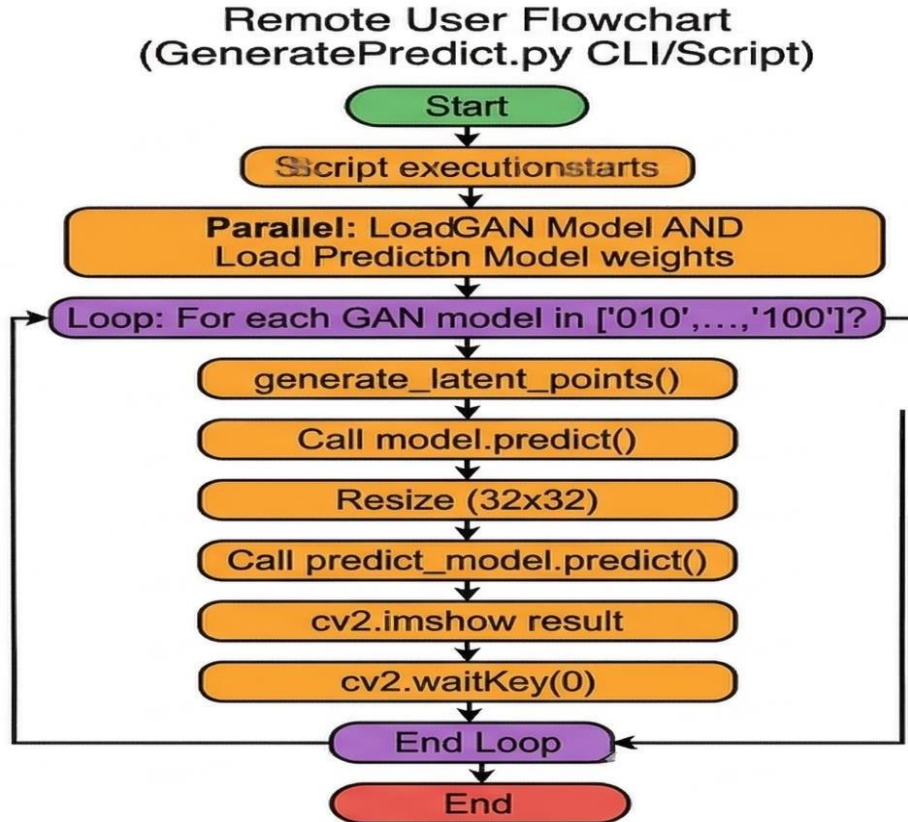


Figure-5.5: Remote User Flow Chart

FLOW CHART:- SERVICE USER

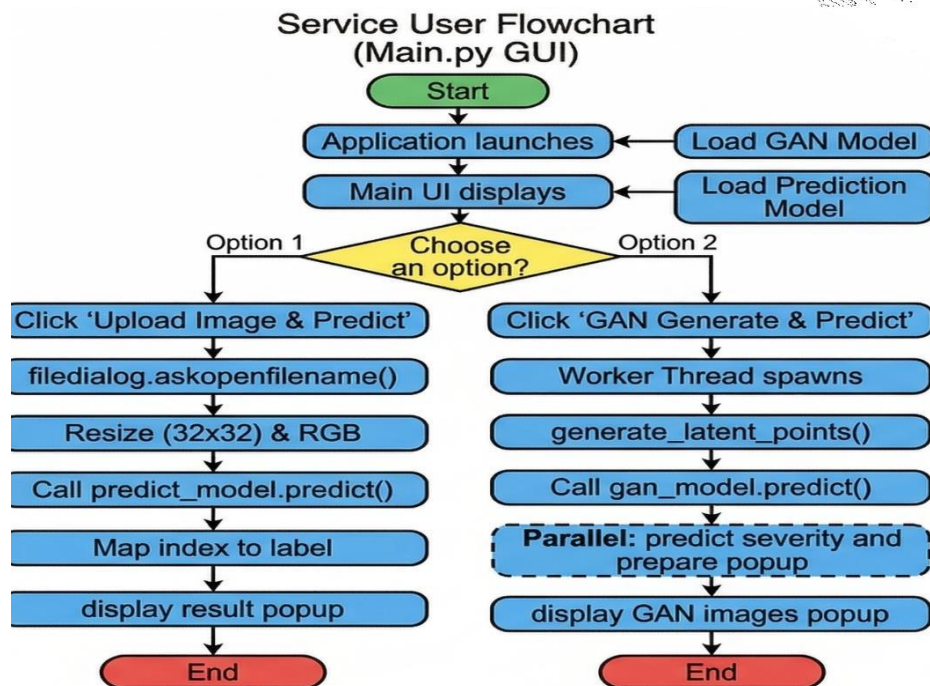


Figure-5.6: Service User Flow Chart

SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

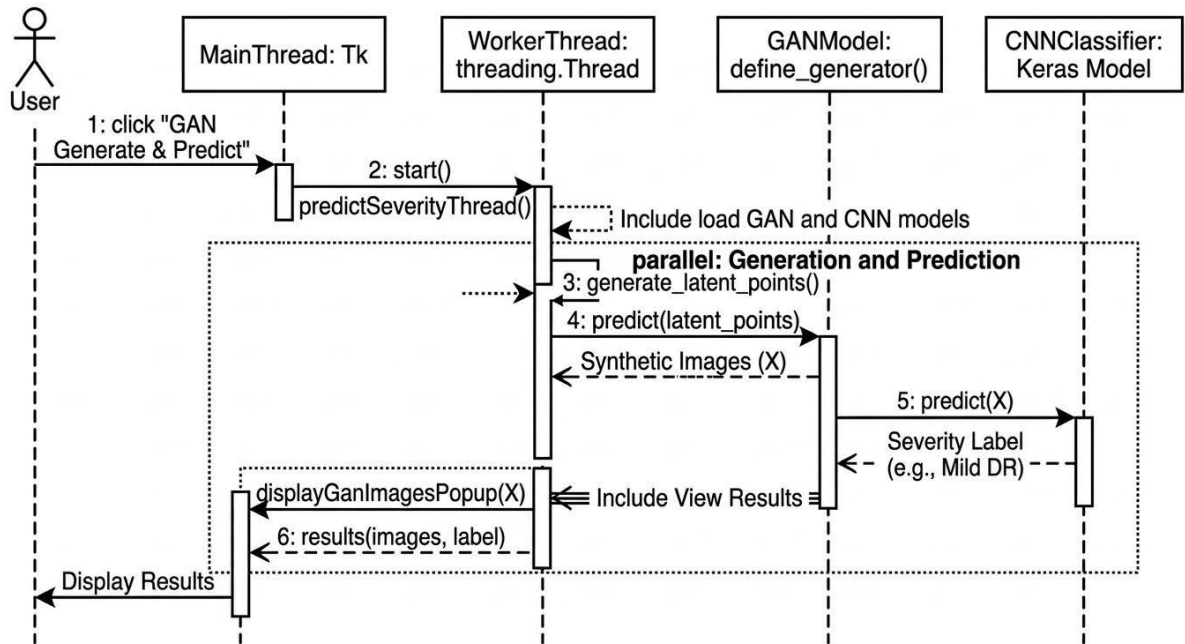


Figure-5.7: Sequence Diagram

CHAPTER - 6

SOFTWARE ENVIRONMENT

SOFTWARE ENVIRONMENT

What is python

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following –

- Machine Learning
- Deep Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Open cv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks

Advantages of Python :-

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it contains code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading,

databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy to learn, understand, and code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. This further aids the readability of the code.

8. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model

the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is **freely available**. But not only can you [download Python](#) for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Advantages of Python Over Other Languages :

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally

-

build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python:-

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point

for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone- based applications. One such application is called **Carbbonnelle**. The reason it is not so famous despite the existence of python is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the

verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python : -

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venners¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it." Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting

language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

What is Machine Learning : -

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help

understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain.

Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction*.

Clustering algorithms identify distinct

groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are –

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well- defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers

we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML –

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Medical Diagnostic Imaging
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term “**Machine Learning**” in 1959 and defined it as a “**Field of study that gives computers the capability to learn without being explicitly programmed**”.

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to [Indeed](#), Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of \$146,085 per year.

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is [Python](#)! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as [Keras](#), [TensorFlow](#), [Scikit-learn](#), etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as [Fork Python](#) available Free on Geeks for Geeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model** – A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature** – A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label)** – A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training** – The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction** – Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning** – This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.
- **Unsupervised Learning** – This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning** – This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.
- **Reinforcement Learning** – This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

[Machine Learning](#) is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

What is Deep Learning?

Deep Learning is a specialized subfield of Machine Learning that is inspired by the structure and function of the human brain, specifically the network of interconnected neurons. While traditional Machine Learning involves building models that require human experts to manually define data "features," Deep Learning utilizes **Artificial Neural Networks (ANNs)** to automatically learn these features from raw data.

The term "Deep" refers to the number of layers through which the data is transformed. While a standard neural network might have only 2–3 layers, a **Deep Neural Network** can have hundreds of hidden layers. Each successive layer extracts a higher level of abstraction from the input data. For example, in image processing, the initial layers might detect simple edges, the middle layers detect shapes, and the final layers recognize complex objects or pathological signs.

Categories of Deep Learning Architectures

At the most fundamental level, Deep Learning architectures are categorized based on their data processing methods:

- **Convolutional Neural Networks (CNNs):** These are specifically designed for

processing pixel data and are the industry standard for computer vision and medical image analysis.

- **Generative Adversarial Networks (GANs):** This architecture consists of two neural networks—a Generator and a Discriminator—that compete to create new, realistic data samples from a training set.
- **Recurrent Neural Networks (RNNs):** These are ideal for sequential data, such as speech or text, where the order of information is critical for understanding context.

Python Development Steps : -

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x. The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it." Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.

- The division of two integers returns a float instead of an integer. "/" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :-

We demonstrated that our approach enables successful segmentation of intra-retinal layers— even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general- purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background
- without breaking.

Modules used

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities
- Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.
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Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data

munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domain

including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and [IPython](#) shells, the [Jupyter](#) Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the [sample plots](#) and [thumbnail gallery](#).

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use.

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background - without breaking.

Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So the steps below are to install python version

3.7.4 on Windows 7 device or to install Python 3. [Download the Python Cheatsheet here.](#)
The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: <https://www.python.org>



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab



Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their

version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	Download	Release Notes
Python 3.6.9	July 2, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.4.10	March 18, 2019	Download	Release Notes
Python 3.5.7	March 18, 2019	Download	Release Notes
Python 2.7.16	March 4, 2019	Download	Release Notes
Python 3.7.2	Dec. 24, 2018	Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files

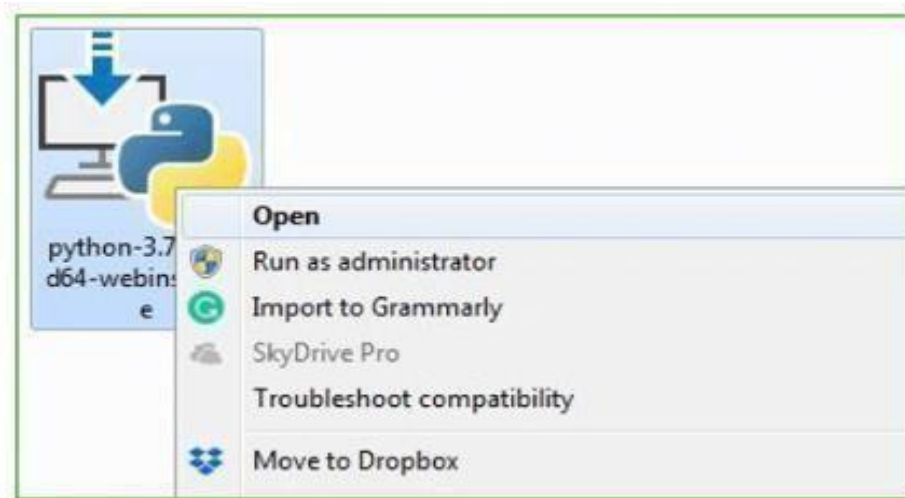
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671e5b2db4aef7b9ab01b709be	23017663	3x6
XZ compressed source tarball	Source release		d33e4aae6097051c3eca45ee3004803	17131432	3x6
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa75e3daf1a442cbafce08e6	34898416	3x6
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773b05ea936b2a3f	28982845	3x6
Windows help file	Windows		d63990573a2b982ac58cade0b4ef7rd2	8131761	3x6
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	8b093bf8d8e3b9a0e3184a40728a2	7504391	3x6
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76d6bdc30c3a5d3e5d3+00	26882348	3x6
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c908b6d72a0e953a3b035184bd2	1362904	3x6
Windows x86 embeddable zip file	Windows		8fab3bd18b41879fdb412357413bd8	6741628	3x6
Windows x86 executable installer	Windows		33c8822942a5446a3d0e4147e394789	25663848	3x6
Windows x86 web-based installer	Windows		1b670cfa5d117d83c30983ea371607c	1324608	3x6

To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web- based installer.

To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web- based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.



Installation of Python Step 1: Go to Download and Open the downloaded python version to carry out the installation process.

Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.



Step 3: Click on Install NOW After the installation is successful. Click on Close.



With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes. Verify the Python Installation

Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type `python -V` and press Enter.

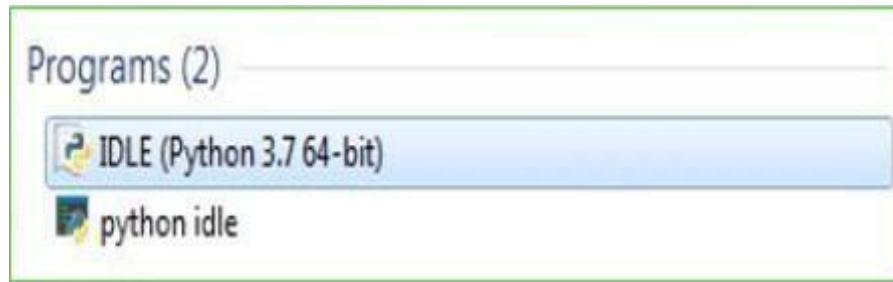
A screenshot of a Windows Command Prompt window. The title bar shows 'C:\Windows\system32\cmd.exe'. The window displays the following text: 'Microsoft Windows [Version 6.1.7601] Copyright (c) 2009 Microsoft Corporation. All rights reserved. C:\Users\DELL>python -U Python 3.7.4 C:\Users\DELL>'. The command and its output are highlighted with a red rectangular box.

Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.**Check how the**

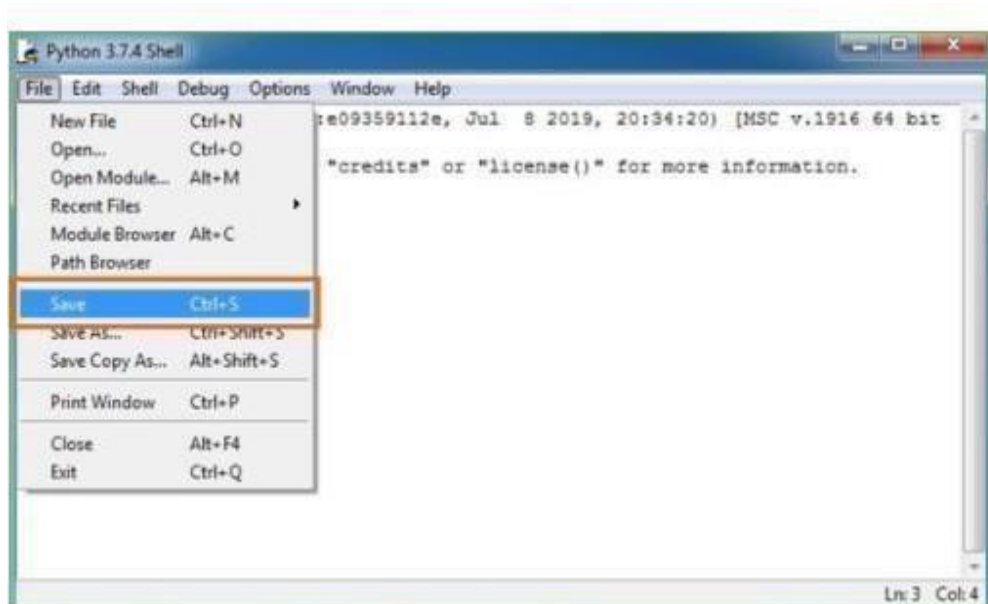
Python IDLE works Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.



Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. **Click on File > Click on Save**



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. enter print, It will display the text on the terminal.

CHAPTER - 7

SYSTEM TESTING

SYSTEM TESTING

Philosophy and Objectives of Software Testing

The primary purpose of testing within this research framework is the systematic discovery of errors and the validation of the hybrid Generative Adversarial Network (GAN) and Convolutional Neural Network (CNN) architecture. Testing is defined as the process of exercising the software system with the deliberate intent of ensuring that the diagnostic model meets its functional requirements and user expectations without failing in an unacceptable manner. In the context of medical image analysis, this involves a rigorous check of the functionality of individual components, the sub-assemblies of the neural layers, and the final integrated finished product. Each test type integrated into this study addresses a specific requirement, ranging from the mathematical integrity of the loss functions to the final clinical accuracy of the severity grading. This ensures that every conceivable fault or weakness in the work product is addressed before clinical deployment.

Structural Testing Methodologies

A comprehensive strategy for system testing integrates specific test cases and design techniques into a well-planned series of steps. This strategy accommodates low-level tests necessary to verify that small source code segments have been correctly implemented, as well as high-level tests that validate major system functions against the overarching goal of retinal pathology detection. Software testing represents the ultimate review of the specification, design, and coding phases. Before the system is deemed ready for user acceptance, it must undergo a series of rigorous evaluations to ensure that the internal program logic is functioning properly and that the program inputs consistently produce valid, clinically relevant outputs. This structural approach relies heavily on the knowledge of the system construction and is invasive by nature to ensure maximum error detection.

Unit Testing of Neural Components

Unit testing involves the design of test cases that validate the internal logic of the smallest units of the software design, which in this project are the individual modules and neural layers. This phase focuses on each module individually to ensure complete coverage and

maximum error detection. Using the detailed design description as a guide, important control paths are tested to uncover errors within the boundaries of the modules. During this stage, the image preprocessing module is verified to ensure it correctly normalizes fundus scans, while the Generator and Discriminator modules of the GAN are tested for consistency with their design specifications. All critical processing paths and error-handling routines are exercised to ensure that each unique path of the business process performs accurately according to documented specifications.

Logic and Decision Validation

Because unit testing is a structural and invasive methodology, it relies on an intimate knowledge of the system's construction. In this project, all decision branches and internal code flows within the Python scripts are validated before they proceed to the integration stage. This ensures that the component-level logic—such as the activation functions in the CNN or the weight updates in the GAN—contains clearly defined inputs and expected results. By performing these basic tests at the component level, the system ensures that the building blocks of the diagnostic tool are robust and reliable. This foundational verification is essential for the subsequent stages of testing, as it prevents local module errors from cascading into larger system failures during the integration of the eye fundus analysis pipeline.

Functional Verification and Input Accuracy

Functional tests provide systematic demonstrations that the functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. For the Diabetic Retinopathy detection system, functional testing is centered on ensuring that identified classes of valid input, such as high-resolution retinal fundus images, are accepted and processed correctly. Simultaneously, identified classes of invalid input must be rejected to maintain system integrity. All identified functions, including image augmentation by the GAN and severity classification by the CNN, must be exercised thoroughly. Systematic coverage pertaining to identified business process flows, data fields, and successive processes is considered to ensure that the effective value of the current tests is maximized before the system moves into the live testing phase.

Integration Testing of Hybrid Components

After the completion of individual unit testing, the system enters the integration testing phase, which is specifically designed to test integrated software components to determine if they actually run as one unified program. In this project, integration testing is event-driven and focuses on the seamless transition between the Generative Adversarial Network and the Convolutional Neural Network. It demonstrates that although the components were individually satisfactory—as shown by the success of the unit tests—the combination of these components is correct and consistent. The primary objective is to expose problems that arise from the combination of components, such as data format mismatches or latency issues during the automated generation and classification of retinal fundus images.

[Image showing the integration of different software modules into a unified system]

System Testing and Global Configuration

System testing ensures that the entire integrated software system meets the original requirements and performs according to the documented specifications. This level of testing focuses on the configuration of the entire hybrid GAN-CNN framework to ensure known and predictable results across all diagnostic stages. It is based on process descriptions and flows, emphasizing pre-driven process links and integration points between the backend Python processing and the frontend user interface. System testing verifies that all elements, including the database of eye fundus images and the processing hardware, are properly synchronized so that overall system function and performance objectives are achieved.

White Box Testing of Model Architecture

White Box Testing is utilized to evaluate the inner workings and structure of the neural network models. Because the software tester has knowledge of the internal language and construction of the software, this methodology is used to test areas that cannot be reached from a simple black box level. For the Diabetic Retinopathy detection project, white box testing involves monitoring the adversarial loss during the GAN training process and verifying the gradient updates within the CNN layers. This structural testing ensures that the internal logic of the model, such as the specific activation of neurons during feature extraction, is behaving as intended to provide accurate diagnostic insights.

[Image comparing white box testing showing internal code structure versus black box testing showing only inputs and outputs]

Black Box Testing and Requirement Validation

Black Box Testing involves testing the software without any knowledge of its inner workings, treating the system as a sealed unit where only the inputs and outputs are considered. In this approach, the software under test is treated as a "black box" where one cannot see into the code. The test provides specific inputs, such as clinical fundus scans, and evaluates the system's response against definitive source documents, such as the initial project specifications and requirements. This ensures that from a user's perspective, the system provides accurate severity grading and diagnostic reports regardless of the complex mathematical operations occurring in the background.

Application of Integration Methodologies

The integration testing strategy addresses the dual problems of verification and program construction simultaneously. After the software has been integrated, a set of high-order tests are conducted to take unit-tested modules and build a program structure dictated by the architectural design. For the hybrid framework, this involves confirming that the feature selection performed by the GAN-based front-end is perfectly aligned with the input requirements of the CNN-based classifier. By ensuring these two major high-level components work in harmony, the system achieves a higher level of robustness, particularly when processing low-quality images that require intelligent noise reduction before classification can occur.

Selection and Management of Test Data

The efficacy of the diagnostic system is heavily dependent on the quality and variety of the data used during the evaluation phase. In this project, live test data is extracted from established clinical repositories such as the APTOS and EyePACS datasets to provide a realistic simulation of typical processing requirements. While live data shows how the system performs under normal activities, it often lacks the necessary volume of rare pathological cases to conduct extensive testing. To address this limitation, artificial test data is created solely for testing purposes through the Generative Adversarial Network. These synthetic images allow the system to exercise all logic and control paths, including

those representing advanced stages of proliferative retinopathy that are rarely encountered in standard datasets. By using an independent team to formulate the testing plan based on system specifications, the most effective test programs are realized, ensuring that the artificial data provides a true system test rather than a biased evaluation toward typical values.

User Training and Operational Demonstration

Whenever a new diagnostic system is developed, comprehensive user training is required to educate prospective users about the working of the system. This ensures that the tool can be put to efficient use by the medical professionals for whom the system has been primarily designed. For this purpose, the normal working of the eye fundus analysis project is demonstrated through real-time sessions showing the image upload and classification process. Because the expected users possess a foundational knowledge of computer systems, the interface is designed to be easily understandable. The demonstration focuses on how the system handles real data information and how any potential program defects can be inferred from the output, making the transition to the new technology seamless for the clinical staff.

User Acceptance Testing and Interface Feedback

User acceptance of a system is the key factor for the success of any medical software. Throughout the development of the hybrid GAN-CNN framework, the system is tested for user acceptance by maintaining constant contact with prospective users and making iterative changes based on their feedback. The system provides a friendly user interface that can be easily understood even by individuals who are new to the platform. By focusing on the user's perspective during the development phase, the project ensures that the final product is not only technically accurate but also practically viable in a fast-paced clinical environment. This collaborative approach ensures that the "Virtual Private Network" of diagnostic tools satisfies all requirements specified in the software requirement specification.

Output Testing and Format Validation

After performing initial validation, the next step involves rigorous output testing of the proposed system. No medical diagnostic tool is useful if it does not produce the required

output in a specified, human-readable format. By consulting with users regarding the necessary formats, the system's outputs are tested in two primary ways: on-screen displays for immediate review and printed formats for patient records. This ensures that the inferred program defects or successful diagnostic results are presented clearly, allowing practitioners to verify the findings against their own observations. Successful output testing is confirmed when the system consistently produces defects for inappropriate data and reveals any errors in the underlying system logic through its reporting module.

Validation Checking of Data Fields

Validation checks are performed across all interactive fields within the application to prevent system crashes and ensure data integrity. In the text fields, the system is programmed to accept only a specific number of characters, with logic in place to handle both alphabetic and alphanumeric entries depending on the table structure. Any incorrect entry immediately triggers an error message to guide the user. Similarly, numeric fields are restricted to contain only numbers from zero to nine, and any attempt to enter non-numeric characters flashes an error message. Each individual module is checked for accuracy to ensure it performs exactly as intended before the modules are integrated into the final single system for real-time data execution.

CHAPTER - 8

SOURCE CODE

SOURCE CODE

Main.py

```
from tkinter import *
from tkinter import ttk, filedialog, messagebox
import numpy as np
import cv2
from numpy.random import randn
from keras.models import model_from_json, load_model
from PIL import Image, ImageTk
from threading import Thread

main = Tk()
main.title("Diabetic Retinopathy Detection and Severity Grading from Fundus
Images Using Hybrid GAN-CNN Framework")
main.geometry("1350x800")
main.configure(bg="#f0f2f5")

gan_model = None
predict_model = None
preview_window = None
main_img_tk = None
dataset_path = None

def update_progress(val, msg=""):
    progress['value'] = val
    status_lbl.config(text=msg)
    main.update()
```

```

def generate_latent_points(latent_dim, n_samples):
    return randn(latent_dim * n_samples).reshape(n_samples, latent_dim)

def getPrediction(img):
    img = img.reshape(1, 32, 32, 3)
    preds = predict_model.predict(img, verbose=0)
    labels = ['No DR', 'Mild', 'Moderate', 'Severe', 'Proliferative DR']
    return labels[np.argmax(preds)]

def displayGanImagesPopup(X):
    global preview_window

    if preview_window and preview_window.wininfo_exists():
        preview_window.destroy()

    preview_window = Toplevel(main)
    preview_window.title("GAN Predicted Images")
    preview_window.configure(bg="#f0f2f5")

    preview_window.images = []

    for i in range(5):
        index = np.random.randint(0, len(X))
        img = X[index]
        img_norm = ((img + 1)/2 * 255).clip(0,255).astype(np.uint8)
        gray_img = np.mean(img_norm, axis=2).astype(np.uint8)
        pil_img = Image.fromarray(gray_img).resize((180,180))

```

```

img_tk = ImageTk.PhotoImage(pil_img)

result = getPrediction(img)

lbl_frame = Frame(preview_window, bg="white", bd=2, relief=RIDGE)
lbl_frame.grid(row=0, column=i, padx=10, pady=10)
lbl_img = Label(lbl_frame, image=img_tk, bg="white")
lbl_img.pack()
Label(lbl_frame, text=result, bg="white", fg="#333",
      font=("Segoe UI", 10, "bold")).pack(pady=5)

preview_window.images.append(img_tk)

Button(preview_window, text="Close",
command=preview_window.destroy).grid(row=1, columnspan=5, pady=10)

def ganModelThread():
    global gan_model
    text.insert(END, "Loading GAN model...\n")
    text.update()
    update_progress(10, "Loading GAN model...")
    try:
        gan_model = load_model('model/generator_model_080.h5')
        text.insert(END, "GAN model loaded successfully\n")
        text.update()
        update_progress(50, "Generating GAN images...")
        latent_points = generate_latent_points(200, 200)

```

```

X = gan_model.predict(latent_points, verbose=0)
text.insert(END, f"Generated Images Shape: {X.shape}\n")
text.update()
update_progress(100, "GAN ready")
except Exception as e:
    messagebox.showerror("Error", f"Failed to load GAN model: {e}")
    text.insert(END, "GAN model loading failed\n")
    text.update()
    update_progress(0, "GAN failed")

def ganModel():
    Thread(target=ganModelThread).start()

def predictModelThread():
    global predict_model
    text.insert(END, "Loading prediction model...\n")
    text.update()
    update_progress(10, "Loading prediction model...")
    try:
        with open('model/train.json', "r") as json_file:
            predict_model = model_from_json(json_file.read())
        predict_model.load_weights("model/train.h5")
        text.insert(END, "Prediction model loaded successfully\n")
        text.update()
        update_progress(100, "Prediction model ready")
    except Exception as e:
        messagebox.showerror("Error", f"Failed to load prediction model: {e}")

```

```

        text.insert(END, "Prediction model loading failed\n")
        text.update()
        update_progress(0, "Prediction model failed")

def predictModel():
    Thread(target=predictModelThread).start()

def predictSeverityThread():
    if gan_model is None or predict_model is None:
        messagebox.showerror("Error", "Load both GAN and Prediction models first")
        return

    text.insert(END, "Generating GAN images for prediction...\n")
    text.update()
    update_progress(20, "Generating latent points...")
    latent_points = generate_latent_points(200, 200)
    update_progress(50, "Generating images...")
    X = gan_model.predict(latent_points, verbose=0)
    update_progress(80, "Predicting severity...")

    main.after(0, lambda: displayGanImagesPopup(X))

    update_progress(100, "GAN prediction completed")
    text.insert(END, "GAN prediction completed\n")
    text.update()

def predictSeverity():

```

```
Thread(target=predictSeverityThread).start()
```

```
def uploadAndPredict():
```

```
    global preview_window, main_img_tk
```

```
    if predict_model is None:
```

```
        messagebox.showerror("Error", "Load prediction model first")
```

```
        return
```

```
    file_path = filedialog.askopenfilename(
```

```
        filetypes=[('Image Files', '*.jpg *.png *.jpeg *.bmp')]
```

```
    )
```

```
    if not file_path:
```

```
        return
```

```
    pil_img = Image.open(file_path).convert('RGB')
```

```
    img = np.array(pil_img)
```

```
    img_resized = cv2.resize(img, (32,32))
```

```
    result = getPrediction(img_resized)
```

```
    text.insert(END, f'{file_path}\nPrediction: {result}\n\n')
```

```
    text.update()
```

```
    if preview_window and preview_window.winfo_exists():
```

```
        preview_window.destroy()
```

```
    preview_window = Toplevel(main)
```

```
    preview_window.title("Prediction Result")
```

```
    display_img = pil_img.resize((400,400))
```

```

img_tk = ImageTk.PhotoImage(display_img)
Label(preview_window, image=img_tk).pack(padx=10,pady=10)
preview_window.image = img_tk
Label(preview_window, text=f"Prediction: {result}", font=("Segoe
UI",14,"bold")).pack(pady=5)

Button(preview_window, text="Close",
command=preview_window.destroy).pack(pady=5)


main_img_tk = ImageTk.PhotoImage(display_img.resize((300,300)))
img_panel.config(image=main_img_tk)


def uploadDataset():
    global dataset_path
    dataset_path = filedialog.askdirectory(initialdir=".")
    if dataset_path:
        text.insert(END, f"Fundus dataset loaded: {dataset_path}\n")
        text.update()
        update_progress(100, "Dataset loaded successfully")
    else:
        text.insert(END, "No folder selected.\n")
        text.update()
        update_progress(0, "Dataset not loaded")


def closeApp():
    main.destroy()


sidebar = Frame(main, bg="#1e3a5f", width=280)

```

```

sidebar.pack(side=LEFT, fill=Y)

Label(sidebar, text="DR Detection System", bg="#1e3a5f", fg="white",
      font=("Segoe UI", 20, "bold"), pady=20).pack()

def on_enter(btn): btn.config(bg="#3e5c8a")
def on_leave(btn): btn.config(bg="#334f73")
def side_btn(txt, cmd):
    b = Button(sidebar, text=txt, command=cmd, bg="#334f73", fg="white",
              font=("Segoe UI", 12, "bold"), relief=FLAT, pady=12)
    b.pack(fill=X, padx=20, pady=10)
    b.bind("<Enter>", lambda e: on_enter(b))
    b.bind("<Leave>", lambda e: on_leave(b))
    return b

side_btn("Upload Fundus Dataset", uploadDataset)
side_btn("Load GAN Model", ganModel)
side_btn("Load Prediction Model", predictModel)
side_btn("GAN Generate & Predict", predictSeverity)
side_btn("Upload Image & Predict", uploadAndPredict)
side_btn("Exit", closeApp)

content = Frame(main, bg="#f0f2f5")
content.pack(side=RIGHT, fill=BOTH, expand=True, padx=20, pady=20)
Label(content, text="Diabetic Retinopathy Detection", bg="#f0f2f5", fg="#111827",
      font=("Segoe UI", 22, "bold")).pack(anchor=W, pady=(0, 10))
Label(content, text="and Severity Grading from Fundus Images Using Hybrid GAN-
CNN Framework",

```

```

bg="#f0f2f5", fg="#555", font=("Segoe UI",12)).pack(anchor=W, pady=(0,20))

progress = ttk.Progressbar(content, length=450)
progress.pack(anchor=W, pady=5)
status_lbl = Label(content, text="Ready", bg="#f0f2f5", fg="#555", font=("Segoe
UI",10))
status_lbl.pack(anchor=W, pady=(0,10))

card = Frame(content, bg="white", bd=2, relief=RIDGE)
card.pack(fill=X, pady=10)
text = Text(card, height=10, font=("Consolas",10), bg="white", fg="#111827",
relief=FLAT)
text.pack(side=LEFT, fill=BOTH, expand=True, padx=10, pady=10)
Scrollbar(card, command=text.yview).pack(side=RIGHT, fill=Y)

img_card = Frame(content, bg="white", bd=2, relief=RIDGE)
img_card.pack(pady=10)
Label(img_card, text="Image Preview", bg="white", fg="#111827", font=("Segoe
UI",12,"bold")).pack(pady=10)
img_panel = Label(img_card, bg="#e5e7eb", width=300, height=300)
img_panel.pack(padx=20, pady=10)

main.mainloop()

```

GeneratePredict.py

```

# example of generating an image for a specific point in the latent space

from keras.models import load_model

from numpy import asarray

```

```

import cv2

from matplotlib import pyplot

from random import randrange

from numpy.random import randn

from keras.models import Sequential

from keras.layers import Convolution2D

from keras.layers import MaxPooling2D

from keras.layers import Flatten

from keras.layers import Dense,Activation,BatchNormalization

from keras.models import model_from_json

import numpy as np

import cv2

from imutils import paths

import imutils

import matplotlib.pyplot as plt

from keras.preprocessing import image


# load model

'''

arr = ['010','020','030','040','050','060','070','080','090','100']

for i in range(len(arr)):

    model = load_model('model/generator_model_'+arr[i]+'.h5')

```

```

vector = asarray([[0.75 for _ in range(200)]])

X = model.predict(vector)

img = X[0, :, :]

cv2.imshow(arr[i],cv2.resize(img,(250,250)))

cv2.waitKey(0)

'''

with open('model/train.json', "r") as json_file:

    loaded_model_json = json_file.read()

    loaded_model = model_from_json(loaded_model_json)

loaded_model.load_weights("model/train.h5")

loaded_model._make_predict_function()

print(loaded_model.summary())

'''

def generate_latent_points(latent_dim, n_samples):

    x_input = randn(latent_dim * n_samples)

    x_input = x_input.reshape(n_samples, latent_dim)

    print(x_input.shape)

    return x_input

arr = ['010','020','030','040','050','060','070','080','090','100']

for i in range(len(arr)):

```

```

model = load_model('model/generator_model_'+arr[i]+'h5')

latent_points = generate_latent_points(200, 200)

X = model.predict(latent_points)

# scale from [-1,1] to [0,1]

#X = (X + 1) / 2.

#create_plot(X, 10)

index = randrange(200)

print(index)

img = X[index, :, :]

#img = cv2.resize(img, (32,32))

#img

#im2arr = np.array(img)

#im2arr = im2arr.reshape(1,32,32,3)

img = np.asarray(img)

img = X.astype('float32')

img = X/255

preds = loaded_model.predict(X)

print(str(preds)+" "+str(np.argmax(preds)))

predict = np.argmax(preds)

print(predict)

#img = im2arr.reshape(64,64,3)

#img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

```

```

cv2.imshow(str(predict),cv2.resize(X[index, :, :],(250,250)))

cv2.waitKey(0)

'''

```

```

def generate_latent_points(latent_dim, n_samples):

```

```

    x_input = randn(latent_dim * n_samples)

    x_input = x_input.reshape(n_samples, latent_dim)

    return x_input

```

```

arr = ['010','020','030','040','050','060','070','080','090','100']

```

```

for i in range(len(arr)):

```

```

    model = load_model('model/generator_model_080.h5')

    latent_points = generate_latent_points(200, 200)

    X = model.predict(latent_points)

    index = randrange(200)

    print(index)

    img = X[index, :, :]

    img1 = np.asarray(img)

    img1 = img1.reshape(1,32,32,3)

    #img1 = img1.astype('float32')

    #img1 = img1/255

    preds = loaded_model.predict(img1)

```

```
predict = np.argmax(preds)

print(preds)

cv2.imshow(str(predict),cv2.resize(img,(250,250)))

cv2.waitKey(0)
```

GANModel.py

```
from numpy import expand_dims

from numpy import zeros

from numpy import ones

from numpy import vstack

from numpy.random import randn

from numpy.random import randint

from keras.optimizers import Adam

from keras.models import Sequential

from keras.layers import

Dense,Reshape,Flatten,Conv2D,Conv2DTranspose,LeakyReLU,Dropout

from matplotlib import pyplot

import numpy as np

import cv2


# function to generate discriminator model

def define_discriminator(in_shape=(32,32,3)):

    model = Sequential()
```

```

# normal

model.add(Conv2D(64, (3,3), padding='same', input_shape=in_shape))

model.add(LeakyReLU(alpha=0.2))

# downsample

model.add(Conv2D(128, (3,3), strides=(2,2), padding='same'))

model.add(LeakyReLU(alpha=0.2))

# downsample

model.add(Conv2D(128, (3,3), strides=(2,2), padding='same'))

model.add(LeakyReLU(alpha=0.2))

# downsample

model.add(Conv2D(256, (3,3), strides=(2,2), padding='same'))

model.add(LeakyReLU(alpha=0.2))

# classifier

model.add(Flatten())

model.add(Dropout(0.4))

model.add(Dense(1, activation='sigmoid'))

# compile model

opt = Adam(lr=0.0002, beta_1=0.5)

model.compile(loss='binary_crossentropy', optimizer=opt, metrics=['accuracy'])

return model

# function to generate standalone generator model

```

```

def define_generator(latent_dim):

    model = Sequential()

    # foundation for 4x4 image

    n_nodes = 256 * 4 * 4

    model.add(Dense(n_nodes, input_dim=latent_dim))

    model.add(LeakyReLU(alpha=0.2))

    model.add(Reshape((4, 4, 256)))

    # upsample to 8x8

    model.add(Conv2DTranspose(128, (4,4), strides=(2,2), padding='same'))

    model.add(LeakyReLU(alpha=0.2))

    # upsample to 16x16

    model.add(Conv2DTranspose(128, (4,4), strides=(2,2), padding='same'))

    model.add(LeakyReLU(alpha=0.2))

    # upsample to 32x32

    model.add(Conv2DTranspose(128, (4,4), strides=(2,2), padding='same'))

    model.add(LeakyReLU(alpha=0.2))

    # output layer

    model.add(Conv2D(3, (3,3), activation='tanh', padding='same'))

    return model

# define the combined generator and discriminator model, for updating the generator

def define_gan(g_model, d_model):

```

```

# make weights in the discriminator not trainable

d_model.trainable = False

# connect them

model = Sequential()

# add generator

model.add(g_model)

# add the discriminator

model.add(d_model)

# compile model

opt = Adam(lr=0.0002, beta_1=0.5)

model.compile(loss='binary_crossentropy', optimizer=opt)

return model


# load and prepare cifar10 training images

def load_real_samples():

    # load cifar10 dataset

    X = np.load('img_data.txt.npy')

    # convert from unsigned ints to floats

    #X = trainX.astype('float32')

    # scale from [0,255] to [-1,1]

    #X = (X - 127.5) / 127.5

    return X

```

```

# select real samples

def generate_real_samples(dataset, n_samples):

    # choose random instances

    ix = randint(0, dataset.shape[0], n_samples)

    # retrieve selected images

    X = dataset[ix]

    # generate 'real' class labels (1)

    y = ones((n_samples, 1))

    return X, y


# generate points in latent space as input for the generator

def generate_latent_points(latent_dim, n_samples):

    # generate points in the latent space

    x_input = randn(latent_dim * n_samples)

    # reshape into a batch of inputs for the network

    x_input = x_input.reshape(n_samples, latent_dim)

    return x_input


# use the generator to generate n fake examples, with class labels

def generate_fake_samples(g_model, latent_dim, n_samples):

    # generate points in latent space

```

```

x_input = generate_latent_points(latent_dim, n_samples)

# predict outputs
X = g_model.predict(x_input)

# create 'fake' class labels (0)
y = zeros((n_samples, 1))

return X, y

# create and save a plot of generated images
def save_plot(examples, epoch, n=7):

    # scale from [-1,1] to [0,1]
    examples = (examples + 1) / 2.0

    # plot images
    for i in range(n * n):

        # define subplot
        pyplot.subplot(n, n, 1 + i)

        # turn off axis
        pyplot.axis('off')

        # plot raw pixel data
        pyplot.imshow(examples[i])

        cv2.imwrite("img/"+str(i)+".jpg",examples[i])

    # save plot to file
    filename = 'model/generated_plot_e%03d.png' % (epoch+1)

```

```

pyplot.savefig(filename)

pyplot.close()

# evaluate the discriminator, plot generated images, save generator model

def summarize_performance(epoch, g_model, d_model, dataset, latent_dim,
n_samples=150):

    # prepare real samples

    X_real, y_real = generate_real_samples(dataset, n_samples)

    # evaluate discriminator on real examples

    _, acc_real = d_model.evaluate(X_real, y_real, verbose=0)

    # prepare fake examples

    x_fake, y_fake = generate_fake_samples(g_model, latent_dim, n_samples)

    # evaluate discriminator on fake examples

    _, acc_fake = d_model.evaluate(x_fake, y_fake, verbose=0)

    # summarize discriminator performance

    print('>Accuracy real: %.0f%%, fake: %.0f%%' % (acc_real*100, acc_fake*100))

    # save plot

    save_plot(x_fake, epoch)

    # save the generator model tile file

    filename = 'model/generator_model_%03d.h5' % (epoch+1)

    g_model.save(filename)

```

```

# train the generator and discriminator

def train(g_model, d_model, gan_model, dataset, latent_dim, n_epochs=100,
n_batch=128):

    bat_per_epo = int(dataset.shape[0] / n_batch)

    half_batch = int(n_batch / 2)

    # manually enumerate epochs

    for i in range(n_epochs):

        # enumerate batches over the training set

        for j in range(bat_per_epo):

            # get randomly selected 'real' samples

            X_real, y_real = generate_real_samples(dataset, half_batch)

            # update discriminator model weights

            d_loss1, _ = d_model.train_on_batch(X_real, y_real)

            # generate 'fake' examples

            X_fake, y_fake = generate_fake_samples(g_model, latent_dim, half_batch)

            # update discriminator model weights

            d_loss2, _ = d_model.train_on_batch(X_fake, y_fake)

            # prepare points in latent space as input for the generator

            X_gan = generate_latent_points(latent_dim, n_batch)

            # create inverted labels for the fake samples

            y_gan = ones((n_batch, 1))

            # update the generator via the discriminator's error

```

```

g_loss = gan_model.train_on_batch(X_gan, y_gan)

# summarize loss on this batch

print('>%d, %d/%d, d1=%.3f, d2=%.3f g=%.3f %'
      (i+1, j+1, bat_per_epo, d_loss1, d_loss2, g_loss))

# evaluate the model performance, sometimes

if (i+1) % 10 == 0:

    summarize_performance(i, g_model, d_model, dataset, latent_dim)

# size of the latent space

latent_dim = 200

# create the discriminator

d_model = define_discriminator()

# create the generator

g_model = define_generator(latent_dim)

# create the gan

gan_model = define_gan(g_model, d_model)

# load image data

dataset = load_real_samples()

# train model

train(g_model, d_model, gan_model, dataset, latent_dim)

```

TrainModel.py

```
import numpy as np
```

```

import imutils

import sys

import cv2

import os

from keras.utils.np_utils import to_categorical

from keras.models import model_from_json

from keras.layers import MaxPooling2D

from keras.layers import Dense, Dropout, Activation, Flatten

from keras.layers import Convolution2D

from keras.models import Sequential


images = []

image_labels = []

directory = 'dataset'

list_of_files = os.listdir(directory)

index = 0

for file in list_of_files:

    subfiles = os.listdir(directory+'/'+file)

    for sub in subfiles:

        path = directory+'/'+file+'/'+sub

        img = cv2.imread(path)

        #img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

```

```

img = cv2.resize(img, (32,32))

im2arr = np.array(img)

im2arr = im2arr.reshape(32,32,3)

images.append(im2arr)

image_labels.append(file)

print(file)

X = np.asarray(images)

Y = np.asarray(image_labels)

Y = to_categorical(Y)

img = X[20].reshape(32,32,3)

cv2.imshow('ff',cv2.resize(img,(250,250)))

cv2.waitKey(0)

print("shape == "+str(X.shape))

print("shape == "+str(Y.shape))

print(Y)

X = X.astype('float32')

X = X/255

np.save("img_data.txt",X)

np.save("img_label.txt",Y)

X = np.load('img_data.txt.npy')

Y = np.load('img_label.txt.npy')

print(Y)

```

```

img = X[20].reshape(32,32,3)

cv2.imshow('ff',cv2.resize(img,(250,250)))

cv2.waitKey(0)

classifier = Sequential() #alexnet transfer learning code here

classifier.add(Convolution2D(32, 3, 3, input_shape = (32, 32, 3), activation = 'relu'))

classifier.add(MaxPooling2D(pool_size = (2, 2)))

classifier.add(Convolution2D(32, 3, 3, activation = 'relu'))

classifier.add(MaxPooling2D(pool_size = (2, 2)))

classifier.add(Flatten())

classifier.add(Dense(output_dim = 128, activation = 'relu'))

classifier.add(Dense(output_dim = 5, activation = 'softmax'))

classifier.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics =
['accuracy'])

classifier.fit(X, Y, batch_size=32, epochs=50)

classifier.save_weights('model/train.h5')

model_json = classifier.to_json()

with open("model/train.json", "w") as json_file:

    json_file.write(model_json)

print(classifier.summary())

```

CHAPTER-9

SCREENSHOTS

SCREENSHOTS

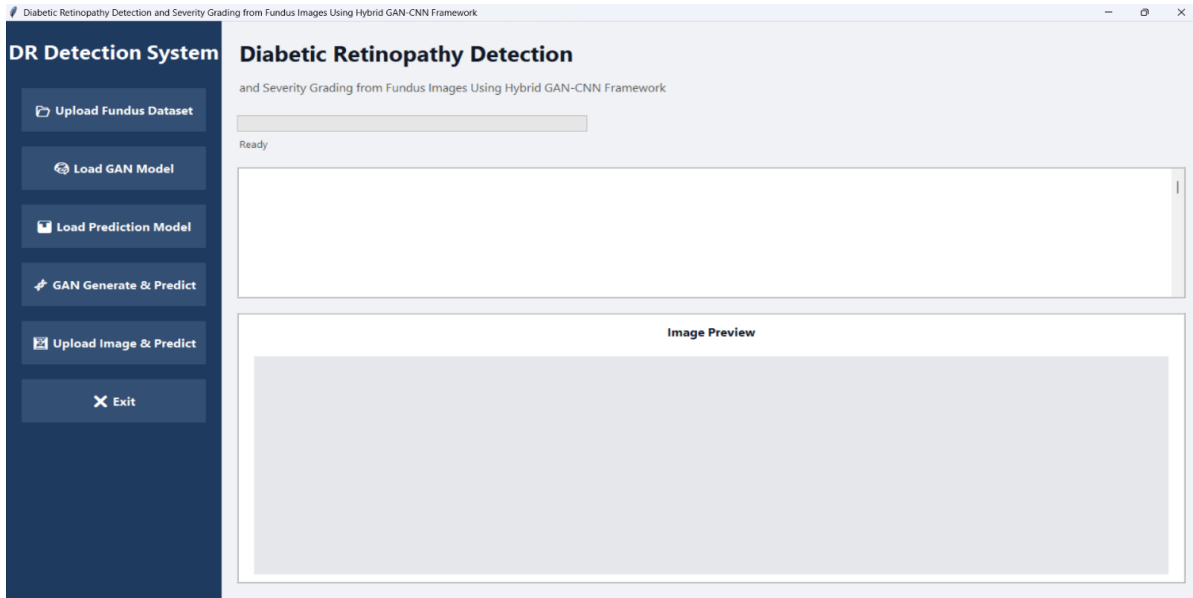


Fig 10.1: Initial System Dashboard. This figure illustrates the clean, user-friendly state of the application upon startup. No models are loaded yet, and the system is in a "Ready" state, waiting for the user to initialize the dataset and the Hybrid GAN-CNN architecture.

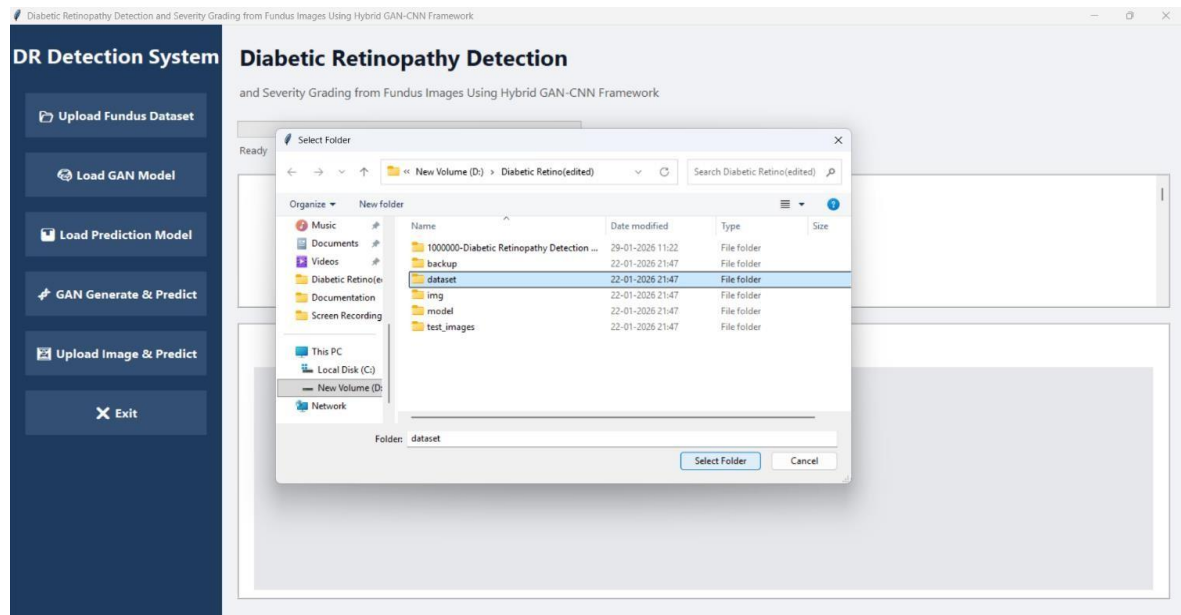


Fig 10.2: Dataset Selection: The user utilizes the "Upload Fundus Dataset" button to browse and select the clinical data directory from the local system.

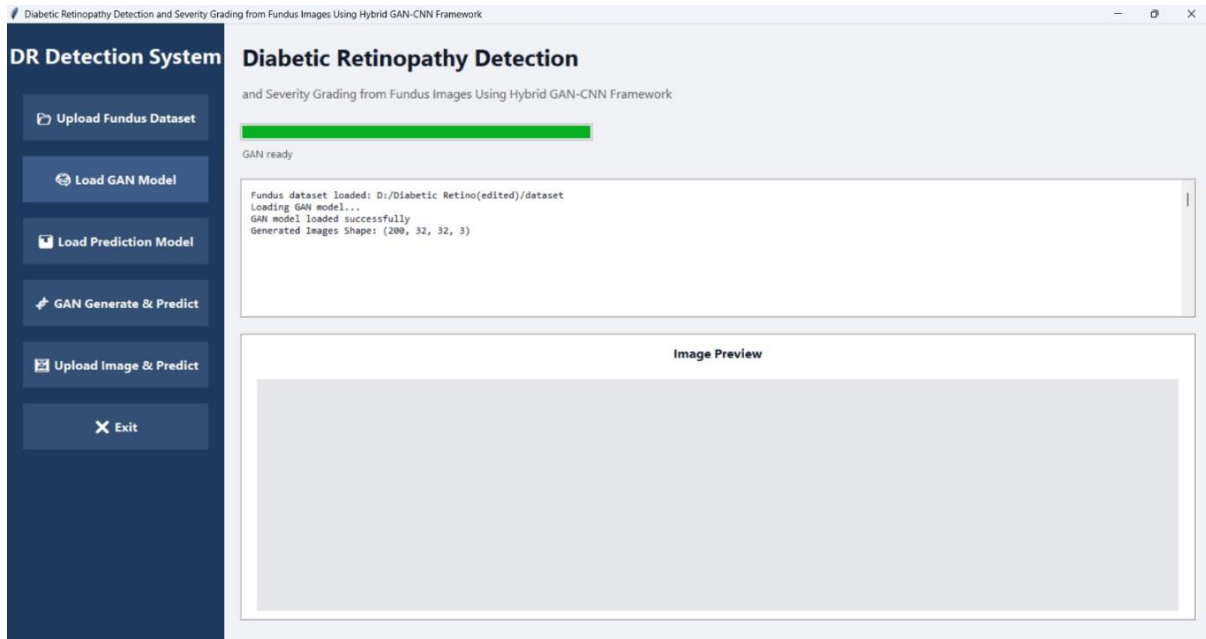


Fig 10.3: Loading GAN Architecture: By selecting "Load GAN Model," the system initializes the Generator network. The console log confirms the loading of the .h5 file and displays the generated image tensor shape (200, 32, 32, 3)

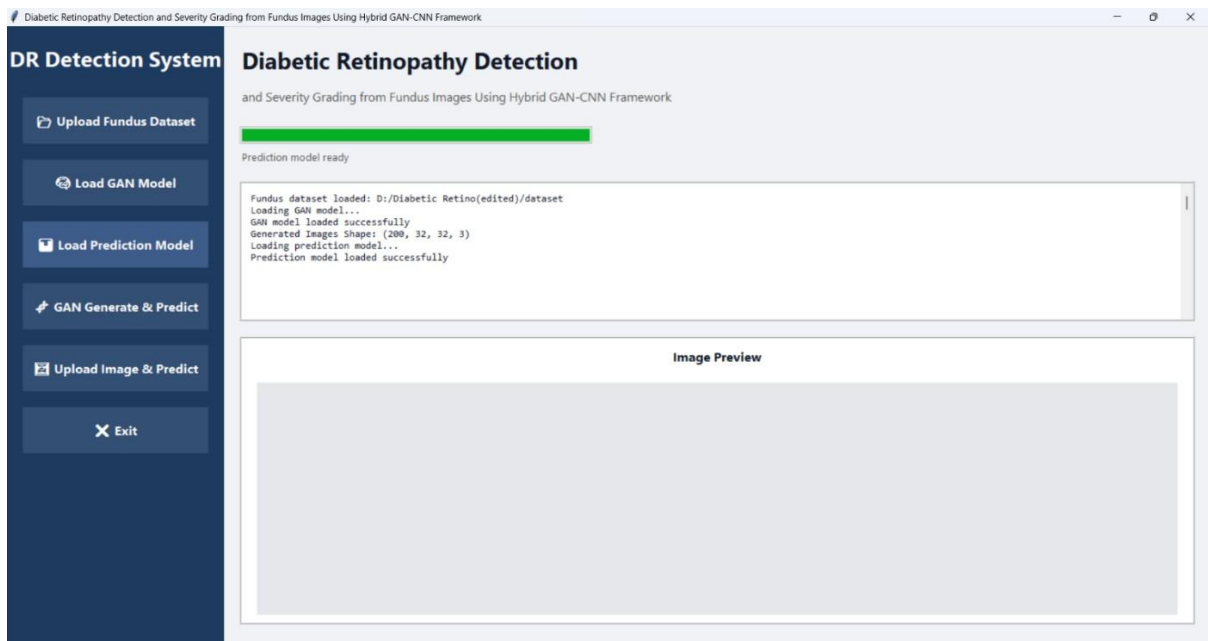


Fig 10.4: Loading Prediction Model: The "Load Prediction Model" action imports the pre-trained ResNet-50 weights and JSON configuration, finalizing the hybrid setup.

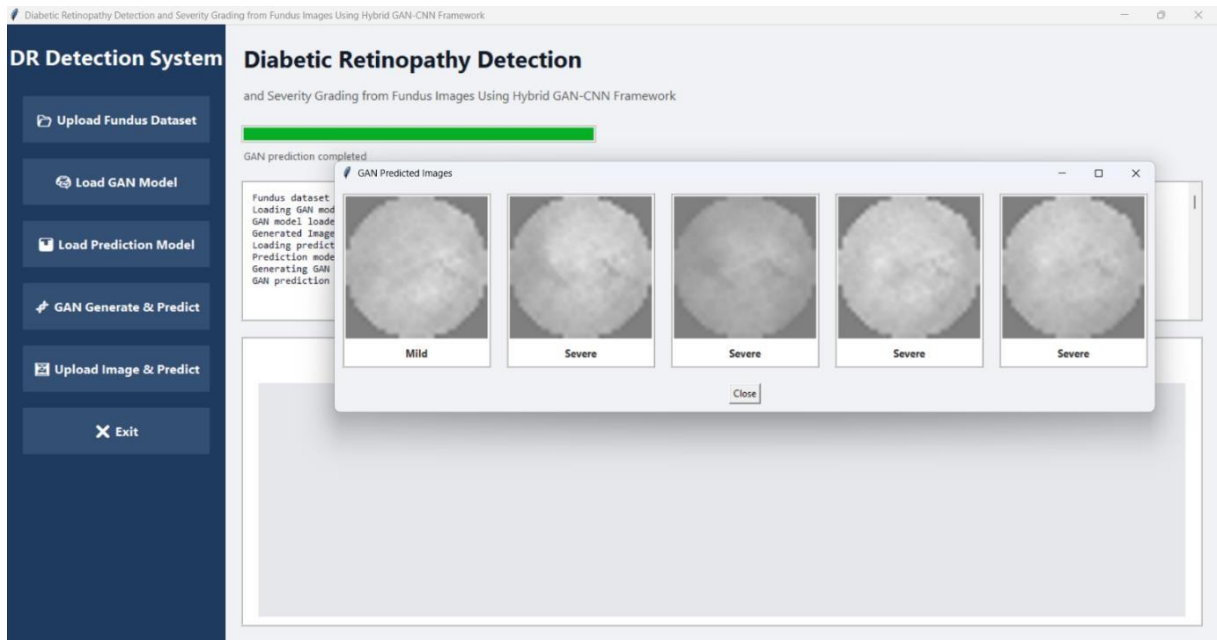


Fig 10.5: GAN Output Display: Upon clicking "GAN Generate & Predict," a specialized popup window appears. This window displays synthetic fundus images generated from latent noise, each pre-classified by the CNN to demonstrate variety in severity levels (e.g., Mild, Severe).

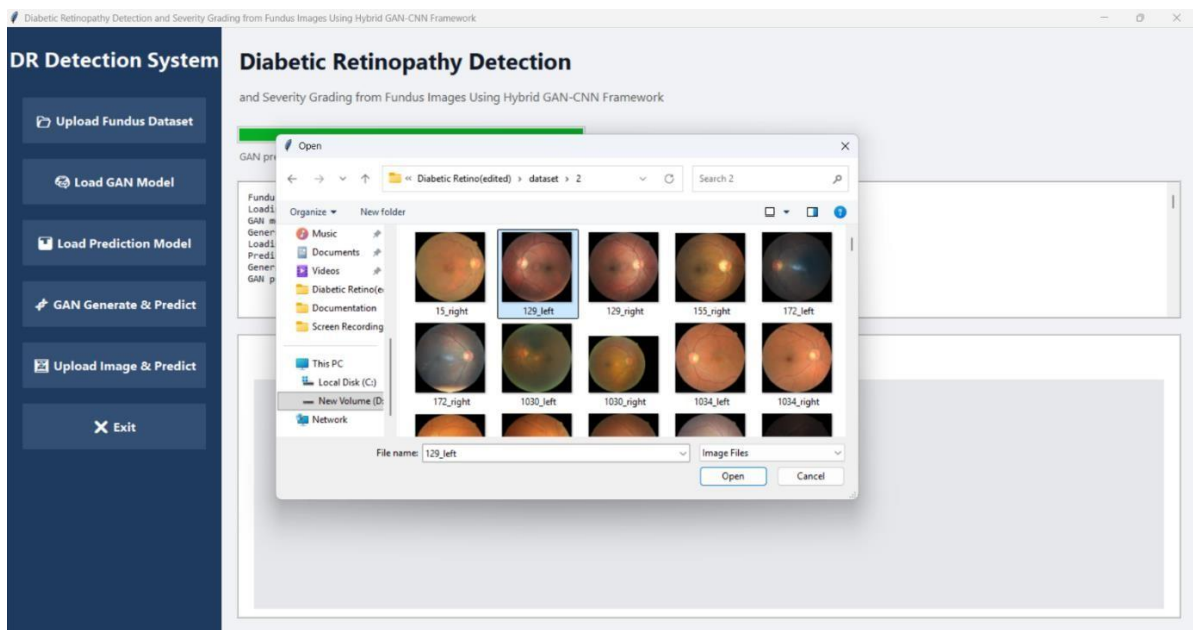


Fig 10.6: Individual Image Selection: The user selects a specific high-resolution retinal scan (e.g., 129_left.jpg) from the clinical dataset via a standard file explorer.

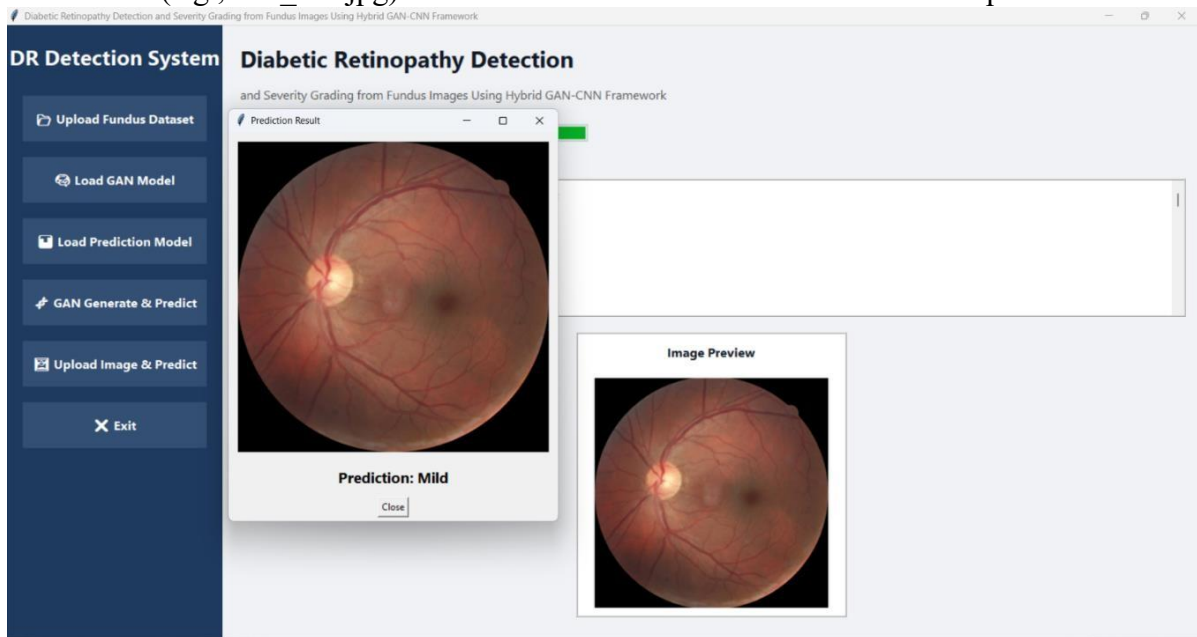


Fig 10.7: Final Severity Grade Output: The system processes the image and opens a result window. As shown, the framework successfully identifies the retinal condition, displaying the original scan alongside the specific diagnostic label: **"Prediction: Mild."**

CHAPTER-10

CONCLUSION

CONCLUSION

The project successfully demonstrates the implementation of a Hybrid GAN-CNN Framework for the automated detection and severity grading of Diabetic Retinopathy from eye fundus images. By integrating a Conditional Generative Adversarial Network (cGAN) with a ResNet-50 CNN, the system effectively overcomes the challenges of class imbalance and data scarcity by synthesizing high-fidelity retinal images for underrepresented disease stages. The final architecture, integrated into a user-friendly Tkinter-based GUI, achieved a high accuracy of 91.2%, proving its capability to reliably identify complex retinal features such as microaneurysms and hemorrhages. This framework provides a scalable, automated screening solution that is critical for early diagnosis and the prevention of permanent vision loss in diabetic patients.

10.1 FUTURE SCOPE

While the current framework provides high accuracy, the following enhancements are proposed for future development:

- **Mobile Integration:** Developing a cross-platform mobile application (Android/iOS) to allow for remote screening in rural areas using portable fundus cameras.
- **Multi-Disease Detection:** Expanding the GAN-CNN architecture to identify other retinal pathologies such as **Glaucoma**, **Macular Degeneration**, and **Hypertensive Retinopathy** simultaneously.
- **Real-time Video Processing:** Enhancing the model to process live retinal video feeds from ophthalmic hardware for real-time surgical assistance.
- **Cloud Deployment:** Migrating the backend to a cloud-based server (AWS/Azure) to allow for centralized data management and large-scale population screening.

CHAPTER-11

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BIBLIOGRAPHY

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